

Mean-payoff parity games in quasipolynomial time

OpenAI

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Abstract

We give a uniform deterministic quasipolynomial-time algorithm for mean-payoff parity games. It computes all vertices from which a player can enforce both nonnegative liminf mean payoff and the parity condition, with arbitrary signed binary rewards and unrestricted binary priorities. The running time is $2^{O((\log(L+2))^2)}$ bit operations, where L is the complete input length.

1 Introduction

A mean-payoff parity game asks a player to satisfy both a long-run reward constraint and a recurring-priority condition. These objectives interact: repeatedly visiting a favorable priority may incur a loss, even when a nonnegative mean reward is achievable on its own. We show that the combined winning region can be computed in deterministic quasipolynomial time in the full binary input length.

An instance consists of a finite directed graph $G = (V, E)$ in which every vertex has an outgoing edge, a partition $V = V_{\text{Max}} \sqcup V_{\text{Min}}$, a reward $r(v) \in \mathbb{Z}$ at each vertex, and a priority $p(v) \in \mathbb{N}$ at each vertex, where $0 \in \mathbb{N}$. The graph with its ownership partition is called an *arena*. The owner of the current vertex chooses the next edge. Strategies are pure: each chooses an outgoing edge after every finite history ending at a vertex owned by that player. A strategy is *positional* if the choice depends only on the current vertex. Max wins an infinite play $\pi = v_0 v_1 \cdots$ precisely when

$$\min\{p(v) : v \text{ occurs infinitely often in } \pi\} \text{ is even,} \quad \text{MP}(\pi) := \liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} r(v_t) \geq 0. \quad (1)$$

Min's winning objective is the complement of this conjunction. Let L be the length of a fixed conventional explicit binary encoding of the graph, ownership, rewards, and priorities. In particular, large reward magnitudes and large priority values are charged by their binary lengths.

Theorem 1.1. *A uniform deterministic algorithm returns exactly all vertices from which Max can force (1) against every Min strategy. For an absolute constant $C > 0$, it uses at most*

$$2^{C(\log_2(L+2))^2}$$

bit operations. The number of distinct priorities and the magnitudes of the rewards are unrestricted.

The algorithmic input is the deterministic mean-payoff theorem of [10, Theorem 1.1], together with its positional-strategy guarantees [10, Corollary 7.1]. We state the exact interface in Proposition 2.1. The reduction from the conjunction to ordinary mean payoff is proved here, including the determinacy and strategy-combination facts used by the recursion. Its number of mean-payoff queries is $2^{O((\log(n+2))^2)}$, where $n = |V|$, and every queried arena has encoding length polynomial in $L + 2$.

The reduction. A *dominion* for a player is a set of vertices within which that player can keep play and win from every start. Following the reduced-precision approach to parity games [11], we compute tentative labels that must be correct on every dominion below a specified size bound for each player. Outside those protected dominions, the labels may be incorrect. A call has at most one recursive child retaining both bounds; every other child halves one bound. Each child removes the parent’s least priority. These two measures give the quasipolynomial call count.

The essential additional issue is how to combine strategies without spoiling mean payoff. A limiting average bound for each uninterrupted play does not itself control the cost of arbitrarily many restarts. We instead construct Max strategies whose every consistent prefix of length t has weight at least $-ct - C_\epsilon$, for every $\epsilon > 0$, with C_ϵ uniform over the opponent’s choices. Using the increasing-round construction of [2, Lemma 7 of the full version], between attempts to visit an even least priority the strategy follows a mean-payoff policy for $1, 2, 3, \dots$ edges. The number of such rounds completed by time T is $O(\sqrt{T})$, so bounded overhead per round has vanishing average. This uniform estimate supports both determinacy and the dominion extraction needed by the recursion.

The two ingredients are separated in the proof: Section 3 establishes the strategy and dominion statements; Section 4 gives the bounded-dominion algorithm; and Section 5 counts its queries and bit operations. The recursion uses the ordinary mean-payoff solver only through its winning-set interface, allowing that solver to be replaced without changing the parity part of the construction.

1.1 Historical context and the role of this reduction

Chatterjee, Thomas A. Henzinger, and Jurdziński introduced mean-payoff parity games to combine qualitative specifications with quantitative performance requirements [6]. They established the existence of optimal strategies and showed that the player enforcing the conjunction may need infinite memory. Bouyer, Markey, Olschewski, and Ummels proved that the opposing player has an optimal positional strategy and gave a simpler recursive value algorithm [2, Theorems 2 and 7]. Thus the strategy asymmetry and the recursive decomposition of these games precede the present work.

Chatterjee and Doyen related mean-payoff parity games to energy parity games, whose accumulated weight must remain above a fixed lower bound, and established membership of the threshold problem in $\text{NP} \cap \text{coNP}$ [4]. Chatterjee, Monika Henzinger, and Svozil improved the threshold running time to $O(n^{d-1}mW)$, for n vertices, m edges, $d \geq 2$ priority levels, and W the maximum of 1 and the absolute weights [5, Theorem 6]. This dependence is polynomial in the numerical weights but exponential in the number of priorities. Daviaud, Jurdziński, and Lazić replaced the latter dependence by a quasipolynomial factor, using strategy decompositions and progress measures combining parity and energy information [7]. Anand, Fijalkow, Goubault-Larrecq, Leroux, and Ohlmann subsequently gave a separating-automaton construction for a related disjunction of parity and mean-payoff objectives with comparable pseudo-quasipolynomial complexity [1]. These bounds retain a numerical weight factor.

For parity games alone, Calude, Jain, Khoussainov, Li, and Stephan obtained a quasipolynomial algorithm [3]; Jurdziński and Lazić developed the succinct progress measures underlying the later mean-payoff parity algorithm [8]. The recursive ancestry relevant here is Parys’s modification of Zielonka’s attractor recursion [11], developed further by Lehtinen, Parys, Schewe, and Wojtczak [9]. Parys’s Algorithm 2 and Lemma 6.1 already use separate size bounds for the two players’ dominions, two sequences of reduced-precision calls surrounding one full-precision call, and the associated dominion-preservation argument. Our algorithm adapts this structure by repeatedly removing vertices where Max loses the ordinary mean-payoff test. The strategy lemmas justify that adaptation for the conjunction.

The strategy construction also has direct predecessors. Increasing payoff phases interspersed with parity visits occur in [6]; the even-priority combination implication is stated in [2, Lemma 5]. The full version of that paper proves it with a phase of length j in round j . The same scheduling appears in the threshold recursion of [5, Appendix C, full version]. Daviaud, Jurdziński, and Lazić also maintain a uniform bound on finite-prefix costs in their strategy-decomposition proof, for their strict mean-payoff convention [7, Sections 2.3–2.4]. Here we make the all- ϵ prefix guarantee explicit for a single strategy at the non-strict threshold and use it to control repeated restarts in the increasing-round construction against history-dependent opponents.

The resulting reduction uses quasipolynomially many ordinary mean-payoff queries of polynomial binary length. Substituting the deterministic mean-payoff algorithm [10, Theorem 1.1 and Corollary 7.1] therefore gives the full binary-length bound. This consequence uses the present query bound together with that companion theorem; the earlier recursive, strategy-combination, and dominion arguments supply its methodological foundations.

2 Mean-payoff input and arena restrictions

We first record what is needed from ordinary mean-payoff games and how winning regions behave under attractor removal. These are the interfaces between the quantitative objective and the recursion.

A finite path $\rho = v_0 \cdots v_t$ has length t and weight

$$\text{wt}(\rho) = \sum_{i=0}^{t-1} r(v_i).$$

Thus assigning edge weight $w(v, u) = r(v)$ preserves every finite prefix weight and the infinite-play payoff. For an arena H , put $W_H = \max_{v \in H} |r(v)|$ when H is nonempty. The mean-payoff objective and both winning objectives in (1) are prefix independent: a finite initial path does not change the recurring priorities or the lower limiting average. A strategy may therefore be restarted at the current vertex with its memory initialized there. All statements about such a restart continue to quantify over arbitrary history-dependent opposing strategies.

2.1 The ordinary mean-payoff interface

Proposition 2.1 (Mean-payoff input). *For a finite nonempty total arena H with signed binary integer weights, its ordinary mean-payoff winning set*

$$U_{\text{mp}}(H) = \{v : \text{Max can force } \liminf \text{ mean payoff at least } 0\}$$

is computable deterministically in $2^{O((\log_2(L_H+2))^2)}$ bit operations, where L_H is its full binary encoding length. There is a single positional Max strategy winning from every vertex of $U_{\text{mp}}(H)$, and a single positional Min strategy forcing strictly negative $\liminf$ mean payoff from every vertex outside it. Both guarantees hold against every history-dependent opposing strategy.

Proof. The winning-set bound is [10, Theorem 1.1]. Its Corollary 7.1 supplies exact values and globally optimal positional strategies. Positional attainment identifies $U_{\text{mp}}(H)$ with the vertices of nonnegative value; at every other vertex Min’s strategy bounds the payoff by its strictly negative value. These assertions use the same $\liminf$ convention as (1). $\square$

Lemma 2.2 (Uniform prefix bound). *If Max wins ordinary mean payoff from every vertex of a nonempty arena H of size N , then a single positional strategy guarantees*

$$\text{wt}(\rho) \geq -(N - 1)W_H$$

for every finite consistent path ρ , from every start.

Proof. Fix the global positional strategy of Proposition 2.1, retaining its chosen edge at each Max vertex and all edges at Min vertices. Every simple directed cycle in this graph has nonnegative weight: otherwise Min could repeat a negative cycle starting on it, contradicting the global guarantee. Delete simple cycles successively from any finite walk. Their nonnegative weights leave a path of at most $N - 1$ edges whose weight is no greater than that of the original walk. Its weight is at least $-(N - 1)W_H$. $\square$

The same cycle argument underlies the winning-policy construction in [10, Proposition 2.2]. Only the winning set in Proposition 2.1 is queried by our algorithm; the strategies are used in its correctness proof.

2.2 Attractors and dominions

We use the standard attractor and dominion properties underlying reduced-precision parity recursion; compare [9, Lemmas 2.1–2.2]. Their short proofs below apply to the present prefix-independent objectives as well.

An *arena restriction* is the induced game on a vertex set that retains at least one outgoing edge at every retained vertex. We allow the empty restriction and identify an arena with its vertex set. A restriction R of H is *closed at player P 's vertices* if every successor in H of a P -vertex in R also lies in R . Write Q for the other player.

For $X \subseteq H$, the attractor $\text{Attr}_P^H(X)$ is the least set containing X that is closed under the following additions: a P -vertex with some successor already in the set, and a Q -vertex with all successors already in the set. Assigning each newly added vertex its stage of entry gives a strategy that decreases this rank until reaching X . Thus P can force a visit to X from its attractor within at most $|H|$ edges.

The complement $H \setminus \text{Attr}_P^H(X)$ is a total restriction closed at P 's vertices. Moreover, a total restriction disjoint from X and closed at P 's vertices is disjoint from the attractor: inductively, no retained P -vertex has a successor in the growing attractor, and each retained Q -vertex has a retained successor outside it.

A *P -dominion* in H is a set D from each of whose vertices P can win the given objective while keeping play in D , against every opposing strategy. Such a set is a total restriction closed at Q 's vertices. Conversely, a total restriction with this closure is a P -dominion exactly when P wins everywhere in its induced game. The empty set is a dominion by convention.

The whole winning region of either player is a dominion for that player, even before determinacy is known. Along a winning strategy, the continuation after any consistent finite history still wins, by prefix independence. Every vertex reached therefore belongs to the same player's winning region. This also shows that the region is closed at the opposing player's vertices.

Lemma 2.3 (Restriction facts). *Let P, Q be opposite players in an arena H .*

- (i) *If D is a P -dominion in H and R is a restriction closed at P 's vertices, then $D \cap R$ is a P -dominion in R .*
- (ii) *If R is closed at P 's vertices, every Q -dominion in R is a Q -dominion in H .*

- (iii) If D is a P -dominion in H and $D \cap X = \emptyset$, then removing $\text{Attr}_Q^H(X)$ leaves D intact and still a P -dominion in the residual arena.

Proof. For (i), use a strategy keeping play in D . At P -vertices its choices remain in R by closure. At Q -vertices every ambient successor stays in D , and the restricted opponent must choose one in R . Hence play remains in the intersection and wins. This also establishes that the intersection is total.

For (ii), Q can choose the edges of its restricted dominion strategy, and P cannot exit R . The same plays and winning condition result in H .

For (iii), D is closed at Q 's vertices and is disjoint from X . The attractor-avoidance observation above shows that it is disjoint from $\text{Attr}_Q^H(X)$. Its winning strategies and the opposing choices within D are therefore retained. $\square$

3 Combining strategies and extracting dominions

We prove determinacy together with a quantitative guarantee for the strategies of Max. This guarantee controls the loss incurred when a strategy is interrupted and later restarted. We then use it to show that an opponent dominion contains a dominion outside an attractor of the least priority. That statement is the structural input to the recursive algorithm.

A fixed strategy σ of Max from a vertex v has the *uniform prefix property* if

$$\text{for every } \epsilon > 0 \text{ there is } C_\epsilon \geq 0 \text{ such that } \text{wt}(\rho) \geq -\epsilon t - C_\epsilon \quad (2)$$

for every σ -consistent path ρ of $t \geq 0$ edges starting at v . The constant is independent of the path and its length. In particular, the strategy is fixed before ϵ is chosen. Dividing the bound by t shows that every consistent infinite play has lower mean payoff at least zero. For any finite collection of strategies satisfying (2), the constants C_ϵ can be chosen in common by taking their maximum.

Each fixed component strategy is restarted with its initial memory. Its prefix bounds therefore apply to the history since the restart, with constants independent of the preceding play.

3.1 Combination across an attractor

When the least priority is odd, repeated visits to it already give Min a win. When it is even, Max must also control the accumulated rewards. The next construction inserts increasingly long portions of a mean-payoff strategy between successive attempts to visit that priority. The qualitative combination argument and its increasing-round construction appear in [2, Lemma 5; full version, Lemma 7]. We give a proof that tracks a uniform bound for all interrupted prefixes.

Lemma 3.1 (Strategy combination). *Let H be a nonempty arena, and let $d \geq 0$ be an integer such that $p(v) \geq d$ for every $v \in H$. Define*

$$P = \begin{cases} \text{Max}, & d \text{ even}, \\ \text{Min}, & d \text{ odd}, \end{cases} \quad S = \{v \in H : p(v) = d\},$$

$$A = \text{Attr}_P^H(S), \quad R = H \setminus A.$$

The set S is allowed to be empty.

1. *If d is odd and Min wins everywhere in the game on R , then Min wins everywhere in H .*
2. *If d is even, Max wins for mean payoff alone everywhere in H , and from every vertex of R Max has a conjunction-winning strategy in R satisfying (2), then from every vertex of H Max has a conjunction-winning strategy satisfying (2).*

Proof. Suppose first that d is odd. On entering R , Min starts a winning strategy for the game on R and follows it until play leaves R . The set R is closed at Min's vertices, so such an exit can only be chosen by Max. In A , Min follows an attracting strategy to S . After reaching S , take one edge, with an arbitrary choice if the vertex belongs to Min, and resume the same rule.

If S is visited infinitely often, its odd priority is the least priority occurring infinitely often, and Min wins. Otherwise play eventually stays in R : each entry into A would force another visit to S in bounded time. The final stay in R follows one winning strategy, so prefix independence again gives a win for Min. The one-edge step after reaching S ensures that successive applications of the rule advance the play.

Now let d be even. By Lemma 2.2, fix a mean-payoff strategy on H whose every finite prefix, from every start, has weight at least $-B$. For each $x \in R$, also fix one conjunction-winning strategy σ_x in R satisfying (2). These choices will remain fixed throughout the construction.

Starting at any vertex of H , Max plays rounds $j = 1, 2, \dots$. Round j has the following parts.

1. Follow the mean-payoff strategy for exactly j edges.
2. If the current vertex is some $x \in R$, start σ_x and follow it for as long as play stays in R . If play never leaves R , this is the remainder of the infinite play.
3. Once play is in A , follow an attracting strategy to S . End the round upon reaching S , counting this as already accomplished if its first vertex in this part lies in S .

The next round starts at the vertex of S just reached. Since R is closed at Max's vertices, any exit in the second part is chosen by Min. If $S = \emptyset$, then $R = H$ and the second part of the first round simply lasts forever.

If infinitely many rounds finish, they give infinitely many visits to S at distinct times: round j contains at least j edges. The least infinitely recurring priority is then the even number d . If only finitely many rounds finish, the unfinished round has an infinite second part, since its first and third parts have bounded length. Its tail follows a conjunction-winning strategy in R . Thus the constructed strategy satisfies parity in either case.

It remains to prove the uniform prefix property; this will also establish the mean-payoff condition. Let W bound the absolute edge weights in H , and fix $\delta > 0$. Choose a common constant C_δ for the finitely many strategies σ_x ; use $C_\delta = 0$ if R is empty. Any internal prefix of the second part, of length t' , has weight at least $-\delta t' - C_\delta$. This applies even if the ambient play subsequently exits R : the internal prefix is a consistent finite path in the total arena R . The exiting edge is counted separately and costs at most W . The attraction then uses at most $|H|$ edges. The four contributions are displayed in Figure 1. Every prefix of a round, including a completed round, therefore has weight at least

$$-\delta t - K_\delta, \quad K_\delta = B + C_\delta + (1 + |H|)W, \quad (3)$$

where t is the length of that prefix.

If q rounds have completed by time T , then $T \geq 1 + 2 + \dots + q$. Including a possible unfinished round, at most $1 + \sqrt{2T}$ rounds contribute to the T -edge prefix ρ_T . Summing (3) gives

$$\text{wt}(\rho_T) \geq -\delta T - K_\delta(1 + \sqrt{2T}).$$

For a given $\epsilon > 0$, set $\delta = \epsilon/2$ and $K = K_{\epsilon/2}$. The elementary inequality

$$K\sqrt{2T} \leq \frac{\epsilon T}{2} + \frac{K^2}{\epsilon}$$

shows that (2) holds with $C_\epsilon = K + K^2/\epsilon$. Only the analysis, not the strategy, depends on ϵ . $\square$

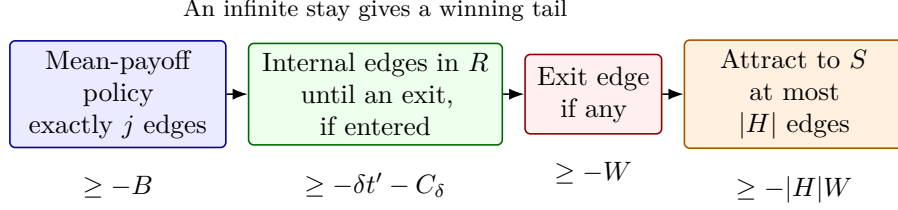


Figure 1: A round in the even-priority construction of Lemma 3.1, with lower bounds for the weights of finite prefixes of its parts. The R -part and its exit edge are omitted when the mean-payoff part ends in A . Block widths are schematic. Every completed round j uses at least j edges, which bounds the number of completed rounds by time T by $O(\sqrt{T})$.

3.2 Determinacy with uniform prefix bounds

We next establish that the strategies required on R in the combination lemma are always available at the winning vertices of Max. Proving this assertion jointly with determinacy avoids any assumption that an arbitrary winning strategy already has uniform prefix bounds. The induction follows the attractor reduction in [5, Algorithm 3 and Appendix C], carrying the quantitative property through each removal.

Theorem 3.2. *In every finite arena, each vertex is winning for exactly one of Max and Min. From each vertex winning for Max, there is a conjunction-winning strategy satisfying (2).*

Proof. We prove both assertions by induction on the number of vertices. They are vacuous in the empty arena. Let H be nonempty and assume the assertions for all smaller arenas.

First observe how a nonempty set of already established winners reduces the induction. Suppose $Y \subseteq H$ is nonempty and every vertex of Y is winning for a player J , with strategies satisfying (2) if $J = \text{Max}$. Every vertex of $Z = \text{Attr}_J^H(Y)$ is also winning for J : attract to Y , then start a winning strategy from the vertex reached. Attraction lasts at most $|H|$ edges. If $J = \text{Max}$, its bounded loss and the finite choice of target strategies preserve (2).

Apply the induction hypothesis to the smaller arena $H \setminus Z$. It is closed at J 's vertices. A strategy there for the opponent of J can therefore keep play in this subarena and remains winning in H . A strategy there for J is followed until any exit to Z , at which point J starts the winning strategy in H associated with the entry vertex. There is at most one such switch: the new strategy is continued on all subsequent histories, even if it later leaves Z . Prefix independence proves that this strategy wins.

When $J = \text{Max}$, the switch also preserves the quantitative property. For each $\epsilon > 0$, the internal prefix in $H \setminus Z$ and the prefix after the switch have lower bounds of the form $-\epsilon t_1 - C_1$ and $-\epsilon t_2 - C_2$. The intervening edge costs at most W , where W bounds the absolute weights in H . Taking the constants in common over the finitely many possible entry vertices gives a bound with slope $-\epsilon$ and constant $C_1 + C_2 + W$. Prefixes before a switch satisfy the first bound. If $J = \text{Min}$, the winning strategies of Max remain entirely in the subarena and retain their prefix bounds without switching. Thus finding such a nonempty Y proves both assertions for H .

Let $d = \min_{v \in H} p(v)$, let P be Max for even d and Min for odd d , and let Q be the other player. Set

$$S = \{v \in H : p(v) = d\}, \quad A = \text{Attr}_P^H(S), \quad R = H \setminus A.$$

If d is even and Max does not win for mean payoff alone everywhere in H , Proposition 2.1 supplies a nonempty set of vertices from which Min can force failure of the mean-payoff condition. These

are conjunction-losing vertices, so they provide Y with $J = \text{Min}$ in the preceding reduction. We may therefore assume that the mean-payoff hypothesis of Lemma 3.1 holds whenever d is even.

Since S is nonempty, R has fewer vertices than H . Apply induction to the game on R . If its Q -winning set is nonempty, these strategies remain winning in H : R is closed at P 's vertices, so Q can keep play there. Their prefix bounds also remain valid when $Q = \text{Max}$. They provide the required set Y with $J = Q$.

In the remaining case, P wins everywhere in R , with the uniform prefix property when $P = \text{Max}$. Lemma 3.1 now proves the assertions for all of H . Finally, a vertex cannot be winning for both players, since their objectives are complementary. $\square$

3.3 A dominion outside the least-priority attractor

We can now turn the combination lemma around. If an opponent dominion had no opponent-winning part outside the relevant attractor, the lemma would give the other player a winning strategy throughout that dominion. For parity games this is the extraction principle of [9, Lemma 2.3]; here the even case also requires the ordinary mean-payoff hypothesis.

Lemma 3.3 (Dominion extraction). *Let H be a nonempty arena and $d \geq 0$ an integer such that $p(v) \geq d$ for all $v \in H$. Let $P = \text{Max}$ if d is even and $P = \text{Min}$ if d is odd, and let Q be the other player. Put*

$$S = \{v \in H : p(v) = d\}, \quad R = H \setminus \text{Attr}_P^H(S).$$

If d is even, assume that Max wins for mean payoff alone everywhere in H . Then every nonempty Q -dominion D in H contains a nonempty Q -dominion of the game on R . The conclusion allows $S = \emptyset$.

Proof. Inside the induced game on D , form

$$R_D = D \setminus \text{Attr}_P^D(D \cap S).$$

Because D is a Q -dominion, it is closed at P 's vertices in H . The attractor complement R_D is closed at P 's vertices in D , hence also in H . It is disjoint from S , so the attractor-avoidance observation in Section 2 gives $R_D \subseteq R$.

If d is even, Max wins for mean payoff alone everywhere in D as well. Indeed, the choices of Max under a winning strategy in H remain available in D , because D is closed at Max 's vertices; Min is restricted to choosing edges in D .

Suppose that the game on R_D has no Q -winning vertex. Theorem 3.2 makes every vertex there winning for P , with the uniform prefix property if $P = \text{Max}$. Apply Lemma 3.1 to the arena D , using the target $D \cap S$ and its residual arena R_D . All its hypotheses have just been verified, so P wins everywhere in D , contrary to the fact that D is a nonempty Q -dominion. This argument includes $R_D = \emptyset$, when its winning-strategy hypothesis is vacuous.

Consequently the Q -winning set in R_D is nonempty. It is a Q -dominion there, and it lifts to a Q -dominion in R by Lemma 2.3, since R_D is closed at P 's vertices. It is contained in D , as required. If $S = \emptyset$, then $R = H$ and $R_D = D$; the same argument applies. $\square$

4 Recursion with bounded-dominion guarantees

The structural results of the preceding section let us adapt the reduced-precision recursion for parity games of Parys [11]. The recursive output is a partition into two labels. At intermediate calls, its

guarantee concerns only dominions below prescribed size bounds. Keeping this distinction explicit is essential: a vertex labelled for a player need not yet be a winning vertex for that player.

Let n be the number of vertices of the original arena. For a subarena H and a pair of integer bounds $\mathbf{b} = (b_{\text{Max}}, b_{\text{Min}}) \in \{0, \dots, n\}^2$, the procedure $\text{Solve}(H, \mathbf{b})$ returns a partition $(\Lambda_{\text{Max}}, \Lambda_{\text{Min}})$ of H . We will prove that, for each $J \in \{\text{Max}, \text{Min}\}$,

$$D \text{ is a } J\text{-dominion in } H \text{ and } |D| \leq b_J \implies D \subseteq \Lambda_J. \quad (4)$$

The call with both bounds equal to n will therefore identify the winning regions exactly. Smaller bounds allow the recursion to make most of its calls with one bound halved.

4.1 The procedure

There are three immediate cases. If $H = \emptyset$, return two empty sets. If $b_{\text{Max}} = 0$, label every vertex Min; this convention also covers the case in which both bounds are zero. If $b_{\text{Max}} > 0$ and $b_{\text{Min}} = 0$, label every vertex Max.

In all remaining cases, define

$$d = \min_{v \in H} p(v), \quad P = \begin{cases} \text{Max}, & d \text{ even,} \\ \text{Min}, & d \text{ odd,} \end{cases} \quad Q \neq P,$$

where Q is the other player. Write

$$b = b_Q, \quad b' = \lfloor b/2 \rfloor, \quad b_P^- = b_P, \quad b_Q^- = b',$$

so that $\mathbf{b}^-$ is obtained from $\mathbf{b}$ by halving the Q -bound. The values $d, P, Q, \mathbf{b}, \mathbf{b}^-$ remain fixed until this call returns. The arena U , initially equal to H , changes by deletions. For each of its current values put

$$S_U = \{v \in U : p(v) = d\}, \quad R_U = U \setminus \text{Attr}_P^U(S_U). \quad (5)$$

In particular, $R_U = U$ if $S_U = \emptyset$.

The subroutine of Proposition 2.1 computes $U_{\text{mp}}(U)$, the ordinary mean-payoff winning set of Max in U . We next define a local procedure $\text{Phase}(U)$, which uses the fixed parameters of its enclosing call and returns a subset of its input arena.

Procedure $\text{Phase}(U)$. While $U \neq \emptyset$, perform the following steps.

1. If d is even, compute $X = U \setminus U_{\text{mp}}(U)$. If $X \neq \emptyset$, replace U by $U \setminus \text{Attr}_Q^U(X)$ and restart the while loop.
2. Form R_U using (5). Call $\text{Solve}(R_U, \mathbf{b}^-)$, and let C be its Q -labelled set. If $C = \emptyset$, return the current U .
3. Replace U by $U \setminus \text{Attr}_Q^U(C)$ and restart the while loop.

If the while loop ends because $U = \emptyset$, return $\emptyset$.

The enclosing call uses two reduced phases separated by a single call with unchanged bounds. The first phase rules out small Q -dominions in R_U . If a Q -dominion of size at most b still survives in U , Lemma 3.3 consequently supplies more than b' of its vertices in a Q -dominion of R_U . The intervening unchanged-bounds call deletes those vertices. At most b' vertices of the original dominion then remain, which is why one final reduced phase suffices.

The precise procedure for $\text{Solve}(H, \mathbf{b})$ is:

- (i) Set $U = H$, and then replace U by $\text{Phase}(U)$.
- (ii) If $U \neq \emptyset$, form R_U and make the single call $\text{Solve}(R_U, \mathbf{b})$ with both bounds unchanged. Let C be its Q -labelled set, and replace U by $U \setminus \text{Attr}_Q^U(C)$.
- (iii) Replace U by $\text{Phase}(U)$ once more.
- (iv) Return $\Lambda_P = U$ and $\Lambda_Q = H \setminus U$.

Calling the phase on the empty arena has no effect. Every deletion is an attractor deletion, so every restriction used here is an arena by the attractor-complement property in Section 2.

We now prove the claimed removal of Q -dominions together with the preservation of P -dominions.

4.2 Termination and correctness

Theorem 4.1 (Bounded-dominion guarantee). *For every subarena H of the original game and every $\mathbf{b} \in \{0, \dots, n\}^2$, the procedure $\text{Solve}(H, \mathbf{b})$ terminates and returns a partition satisfying (4) for both players.*

Proof. We induct on the number of distinct priorities in H . The empty-arena and zero-bound cases terminate immediately and have the stated guarantee: the only dominion subject to a zero size bound is empty, and the other player receives the whole arena.

Consider a call with $H \neq \emptyset$ and both bounds positive. Its fixed value d occurs in H . Every child arena R_U excludes all vertices of priority d and is a subset of H . It therefore has strictly fewer distinct priorities than H , so the induction hypothesis applies to every child. This remains true if d has disappeared from U before the child is formed: the strict decrease is relative to the parent's input H , even when $R_U = U$. Each continuation of a phase loop deletes a nonempty set. Thus its children terminate by induction, its loop can continue at most $|H|$ times, and the enclosing call terminates.

Preservation of P -dominions. Fix a P -dominion D in H with $|D| \leq b_P$. We show that every deletion leaves all of D in the current arena and preserves it there as a P -dominion. Suppose this holds immediately before a deletion. For a deletion in the mean-payoff step, $P = \text{Max}$. A winning strategy that stays in D also wins for mean payoff alone in U , so $D \cap X = \emptyset$. Since D is closed at Q 's vertices, Lemma 2.3 shows that $\text{Attr}_Q^U(X)$ misses D and preserves its dominion property in the residual arena.

For a deletion obtained from a recursive call, the restriction R_U is closed at P 's vertices. The same lemma therefore makes $D \cap R_U$ a P -dominion in R_U . Its size is at most b_P , and both kinds of child call retain this bound for P . By the induction hypothesis, the child labels every vertex of $D \cap R_U$ for P . Its Q -labelled set $C \subseteq R_U$ is consequently disjoint from D . Again $\text{Attr}_Q^U(C)$ misses D and preserves it as a dominion. This argument applies to deletions in either phase and in the unchanged-bounds stage. Hence D lies in the returned set Λ_P . In particular, this reasoning uses only the child's guarantee on small dominions; it does not require that every vertex of C actually be winning for Q .

Removal of Q -dominions. Fix a Q -dominion D in H with $|D| \leq b$. Every current arena is obtained from the preceding one by removing a Q -attractor. Each such restriction is closed at Q 's vertices. Applying the intersection part of Lemma 2.3 after each deletion gives

$$D \cap U \text{ is a } Q\text{-dominion in the current arena } U. \quad (6)$$

Unlike the preceding preservation argument, this assertion allows some vertices of D to have been deleted.

Suppose a phase returns a nonempty arena U . It did so because its last reduced-bound child returned no Q -labelled vertices. The induction hypothesis then implies that

$$R_U \text{ contains no nonempty } Q\text{-dominion of size at most } b'. \quad (7)$$

Moreover, when d is even, the mean-payoff step immediately before that child certified $U_{\text{mp}}(U) = U$. Thus, at a nonempty phase return, all hypotheses of Lemma 3.3 hold in U : every priority is at least d , and the required mean-payoff condition holds when d is even. No occurrence of priority d in the current U is required.

Let U_1 be the arena returned by the first phase. If $D \cap U_1$ is nonempty, it is a Q -dominion there by (6). Lemma 3.3 gives a nonempty Q -dominion E of the game on R_{U_1} such that

$$E \subseteq D \cap U_1, \quad b' < |E| \leq b.$$

Here the strict lower bound follows from (7). The unchanged-bounds child has Q -bound b , so it labels all of E for Q . Its subsequent attractor deletion therefore removes at least $b' + 1$ vertices of D , leaving at most $|D| - (b' + 1) \leq b - b' - 1 \leq b'$ vertices. If $D \cap U_1$ was already empty, no vertices of D remain. Thus, in either case, the arena U at the start of the final phase satisfies

$$|D \cap U| \leq b'. \quad (8)$$

Let U_2 be the arena returned by the final phase. If $D \cap U_2$ were nonempty, it would be a Q -dominion in U_2 of size at most b' by (6) and (8). Applying Lemma 3.3 once more would produce a nonempty Q -dominion in R_{U_2} of size at most b' , contradicting (7). Therefore $D \cap U_2 = \emptyset$, and every vertex of D receives label Q . The two arguments prove the induction step for both players. $\square$

To solve the original arena G , call $\text{Solve}(G, (n, n))$ and return its Max-labelled set. By Theorem 3.2, the two winning regions partition G , and the winning-region observation in Section 2 makes each a dominion. Their sizes are at most n , so Theorem 4.1 labels both regions correctly. It remains to bound the cost of computing this partition.

5 Complexity in the full binary input length

We first bound the recursion independently of the cost of solving an ordinary mean-payoff game. This separates the combinatorial reduction from the numerical subroutine and shows exactly where the full binary encoding enters the running time.

Proposition 5.1 (Cost of the reduction). *For an arena with $n \geq 1$ vertices and complete binary input length L , the call $\text{Solve}(G, (n, n))$ makes*

$$2^{O((\log_2(n+2))^2)}$$

calls to the ordinary mean-payoff winning-set subroutine. Each such call has an explicit input of length $(L + 2)^{O(1)}$. Apart from those subroutine calls, the computation takes

$$2^{O((\log_2(n+2))^2)}(L + 2)^{O(1)}$$

bit operations. All constants are absolute.

Proof. Consider the tree of calls to Solve, counting its immediate base cases as nodes. Every child omits the minimum priority of its parent's input arena. Thus a path has at most n edges, independently of the numerical values of the priorities.

Each node has at most one child with both size bounds unchanged. Every other child has one bound halved, with rounding down. In either phase, a child is followed by a nonempty deletion unless that phase returns. Each phase therefore makes at most $n + 1$ child calls, giving at most

$$M = 2(n + 1)$$

children with a halved bound at a node. Along a path, either one of the two size bounds can be halved at most $1 + \lfloor \log_2 n \rfloor$ times before reaching zero, where the recursion stops. Consequently, there are at most

$$h = 2(1 + \lfloor \log_2 n \rfloor)$$

halving edges on any path.

The resulting path count is the same combinatorial saving that underlies reduced-precision parity recursion [11]. For a path of length ℓ with j halving edges, choose their positions and, at each one, the child index among at most M halving children. At every remaining position there is at most one available unchanged-bounds child. This gives at most $\binom{\ell}{j} M^j$ paths of that type. As each node is specified by its root-to-node path, the total number of calls N_{rec} satisfies

$$\begin{aligned} N_{\text{rec}} &\leq \sum_{\ell=0}^n \sum_{j=0}^{\min(\ell, h)} \binom{\ell}{j} M^j \\ &\leq (n + 1)(h + 1)(nM)^h = 2^{O((\log_2(n+2))^2)}. \end{aligned} \tag{9}$$

The second inequality uses $\binom{\ell}{j} \leq n^j$ and $nM \geq 1$; it also covers $n = 1$.

We next count the nonrecursive work at a node. A phase iteration that continues deletes at least one vertex. The two phases together therefore make at most $n + 2$ iterations, allowing one final return iteration for each phase. In particular, they make $O(n + 1)$ ordinary mean-payoff calls. Attractor computations, construction of induced subarenas, scans for minimum priorities, and updates of the two size bounds all take polynomially many bit operations in $L + 2$. No step iterates through the integers between two successive priority values: minimum priorities and equality tests are computed directly from their binary encodings.

All mean-payoff instances use induced subarenas of the original graph, with the weights $w(v, u) = r(v)$. Renumbering retained vertices uses only polynomial space and time. Even if the reward $r(v)$ is copied explicitly onto every retained outgoing edge, the total number of weight bits is at most the number of edges times the largest original reward bit length, hence polynomial in $L + 2$. The rest of each subroutine input is also polynomial in $L + 2$. This proves the claimed input-length bound without a dependence on the numerical magnitude of a reward. Combining these per-node bounds with (9) proves the proposition. $\square$

Proof of Theorem 1.1. The empty arena has an empty winning set and is handled immediately. For a nonempty arena, the output of Solve($G, (n, n)$) is exact by Theorem 4.1 and the winning-region argument at the end of Section 4.

Proposition 2.1 implements each ordinary mean-payoff call of input length L' in $2^{O((\log_2(L'+2))^2)}$ bit operations, uniformly and deterministically. By Proposition 5.1, there is an absolute constant a such that $L' \leq (L + 2)^a$ for every call, after increasing a if necessary. Its cost is therefore

$2^{O((\log_2(L+2))^2)}$. The number of calls and all remaining work have the same bound, since n is polynomially bounded by the explicit input length. Their product is again

$$2^{O((\log_2(L+2))^2)}.$$

All graph operations and recursion choices admit deterministic implementations, so the complete procedure has the stated uniform deterministic bit complexity. In particular, the constants do not depend on the number of priorities or on the numerical sizes of the binary rewards and priorities. $\square$

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